Ali Abbas

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Education

University of British Columbia Sept. 2023 – April 2028
B.Sc. Data Science – Outstanding Intl. Student Award *Vancouver, BC*

Experience

Software Engineer @ ESGTree June 2026 – Dec 2026
- Built a lender-facing web dashboard in **React** and **Leaflet** rendering 75+ farm operations on an interactive map of Canada, with per-farm detail views (emissions breakdown, crop rotation history, soil profile) and **XLSX** export.
- Shipped a reusable **Python** package that computes crop rotation nitrogen credits and regulatory compliance verdicts from field-level cropping history, benchmarked against published provincial agronomic and cost-of-production data.
- Engineered an **ETL** pipeline integrating four public data sources (satellite crop inventory rasters, national soil polygons, Statistics Canada yield tables, and a federal emissions model) into a normalized field-level schema powering downstream estimates.
- Expanding scope from Sept. 2026 to the platform's **financed emissions** module for a Desjardins pilot deployment, enabling commercial borrowers to self-report operational emissions for **Scope 3 Category 15** disclosure.

Software Developer @ UBC MINT Engineering Design Team Sep 2024 – Present
- Built a real-time streaming visualization app in Next.js using WebSockets to render live EEG data at **256 msg/s**, with low-latency charts (Recharts) and a component library built on shadcn/ui**,** Radix UI, and Tailwind CSS.
- Refactored two duplicated EEG and EMG signal-validation scripts into a single **Python** module with backward-compatible entry points, preserving both existing call sites and validating every check against fault cases.
- Implemented an **EMG preprocessing pipeline** (**SciPy**) with bandpass and notch filtering, RMS envelope extraction, and MVC normalization, feeding a node-based signal processing backend over **Lab Streaming Layer**.

Full-Stack Developer @ Electricity and Water Authority, Bahrain Jul 2025 – Aug 2025
- Built a full-stack ticket management system (Jira-like) with **React**, **Node.js/Express**, and **MongoDB**; designed **RESTful API** endpoints and collections for projects, tickets, and sprints.
- Integrated **Slack API** webhooks for real-time updates and used **OpenAI (GenAI)** to auto-generate project status summaries and stand-up notes.
- Collaborated using **Agile/Scrum** practices (stand-ups, sprint goals) and authored concise project documentation/runbooks.

Mobile QA Engineer @ Qenta Jul 2024 – Aug 2024
- Wrote automated **UI and unit tests** in **Swift** using **XCTest** for a production **iOS** payments application, covering checkout and transaction flows across multiple device and OS-version configurations.
- Identified and reproduced a **payment gateway crash** that surfaced only on specific iOS versions, isolating the version-dependent conditions and documenting reproduction steps for the engineering team.
- Authored structured **test cases** and defect reports from product requirements, working directly with the engineering manager to triage and verify fixes.

Technical Projects

Engine Fault Detection Pipeline | *Python, Docker, AWS S3, Streamlit, LangChain*
- Built an end-to-end **data pipeline** (**Python**, **Pandas**) with modular stages for schema validation, rolling-window feature engineering, and fault classification, persisting versioned artifacts to **AWS S3** behind a **swappable storage layer** (local or cloud through one interface).
- Containerized the pipeline with **Docker** and added **automated tests**, including a guardrail test enforcing no data leakage across the train/test split, for reproducible runs across environments.
- Shipped an interactive **Streamlit** dashboard that ranks engines by predicted failure risk over a web UI; the backing **Random Forest** model reached **0.88 F1** and **0.93 recall** on held-out engines.

Technical Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, C++, R
Frameworks & Web: React, Next.js, Node.js/Express, REST APIs, WebSockets, Leaflet, Tailwind CSS
Data & ML: Pandas, NumPy, Scikit-Learn, PostgreSQL, MongoDB, Oracle SQL, ETL
Tools: Docker, AWS (S3), Git/GitHub, pytest, CI/CD, LaTeX